

Experiment 11 :

Implement a simple top-down parser for context-free grammars using python.

Code:

```
import nltk

from nltk import CFG

from nltk.parse import RecursiveDescentParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'boy' | 'ball'
V -> 'kicked'
""")

parser = RecursiveDescentParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

found = False

for tree in parser.parse(sentence):

    print(tree)

    found = True

if not found:

    print("Invalid Sentence")
```

Output

```
grammar = CFG.fromstring("""
S -> NP VP
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Enter a sentence: the boy kicked the ball
(S (NP (Det the) (N boy)) (VP (V kicked) (NP (Det the) (N ball)))))
```